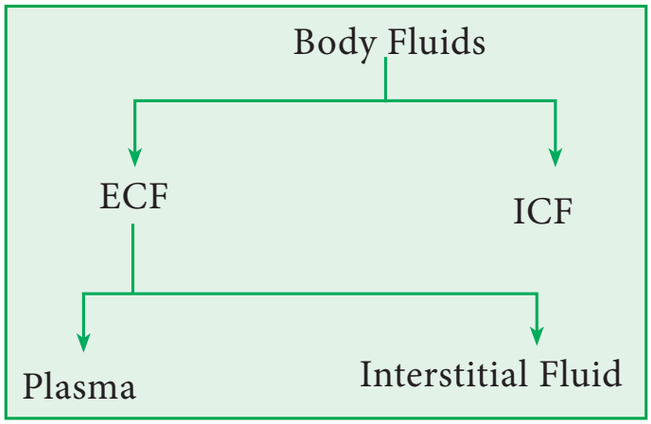

Flowchart – Classification of Body fluids

Body fluids are watery solution of dissolved substances such as oxygen, nutrients and wastes etc. Depending upon their location (compartment), they are of two types, namely, intracellular fluid (ICF) and extracellular fluid (ECF). Intracellular fluid is the fluid present within all the cells of the body. Intracellular fluid is two thirds of TBW i.e. 40% of body weight. The major cations of ICF are K^+ and Mg^{2+}. The major anions are proteins and organic phosphates.

Collectively, the fluid present in the blood and in the spaces surrounding cells is called extracellular fluid (ECF), that is, all the fluid that is outside of cells. The ECF is one thirds of TBW i.e. 20% of body weight. The major cation is Na^+. The major anions are Cl^- and HCO_3^-. The ECF is comprised of plasma (1/4th of ECF) and interstitial fluid (3/4th of ECF).

The interstitial fluid (tissue fluid) lies around and between cells. Its composition is the same as that of plasma except that it lacks larger proteins. Thus, interstitial fluid is an ultrafiltrate of plasma. Cerebrospinal fluid and lymph are examples for interstitial fluid.

Cerebrospinal fluid (CSF)

The cavities of the brain (ventricles), the spinal cord and subarachnoid region is filled with CSF. The total volume of CSF is 100 – 150 ml. It is a clear, transparent and colorless fluid. It has similar pH as that of blood (7.20 to 7.40 i.e. slightly alkaline). It protects the brain and spinal cord from shocks and maintains a uniform pressure on the nervous structures. It acts as a reservoir to regulate the contents of the cranium. To a limited extent, it acts as a medium for nutrient exchange in the nervous system.

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